

(a) the magnitude of the impulse exerted by P on Q ,

(2)

(b) the value of m .

(3)



(a) show that $R=860$,

(3)

(b) find the tension in the tow-bar.

(3)

4.

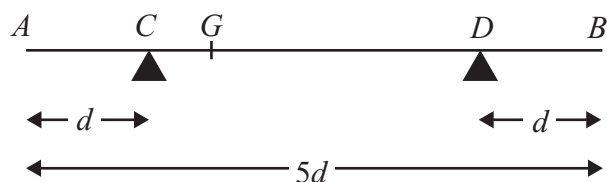


Figure 1

A non-uniform rod AB , of mass m and length $5d$, rests horizontally in equilibrium on two supports at C and D , where $AC = DB = d$, as shown in Figure 1. The centre of mass of the rod is at the point G . A particle of mass $\frac{5}{2}m$ is placed on the rod at B and the rod is on the point of tipping about D .

- (a) Show that $GD = \frac{5}{2}d$.

The particle is moved from B to the mid-point of the rod and the rod remains in equilibrium.

- (b) Find the magnitude of the normal reaction between the support at D and the rod. (5)





8.

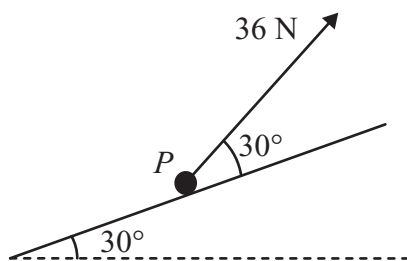


Figure 2

A particle P of mass 4 kg is moving up a fixed rough plane at a constant speed of 16 m s^{-1} under the action of a force of magnitude 36 N . The plane is inclined at 30° to the horizontal. The force acts in the vertical plane containing the line of greatest slope of the plane through P , and acts at 30° to the inclined plane, as shown in Figure 2. The coefficient of friction between P and the plane is μ . Find

- (a) the magnitude of the normal reaction between P and the plane, (4)
- (b) the value of μ . (5)

The force of magnitude 36 N is removed.

- (c) Find the distance that P travels between the instant when the force is removed and the instant when it comes to rest. (5)

